pfi

# Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

# Case Study: The First RAG Pipeline

WEEK 1 · DAY 5

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# Today: the first RAG pipeline

## The parts, joined

| | |
|---|---|
| **Chunks** | 1,158 at 512/64 (day 3) |
| **Index** | MiniLM, FAISS exact (day 4) |
| **Retrieval** | dense, k = 3 |
| **Questions** | the five arc questions |
| **Answers** | 12 saved; replayed, no key |

## Today

1. Every setting in one record
2. The retriever
3. The prompt
4. One answer
5. The local route, on a T4
6. Case study: fifteen answers, graded
7. What the pipeline shows, and where it fails

**Every answer cites a chunk or says NOT IN CORPUS; every count says which rule it uses.**

# Every setting in one record

**RagSettings(index="faiss").settings_line(): computed, never typed**

```
chunker=recursive 512/64, embedder=minilm@1110a24, index=faiss exact, retrieve=dense k=3, rerank=none, expand=0, model=gpt-4.1-mini, temperature=0, top_p=1, max_tokens=300
```

## 6579143b53aa…

settings_hash(), k = 3

## 9b08e87b0ea7…

settings_hash(), k = 5

## frozen

a new setting is a new record

```
FrozenInstanceError: cannot assign to field 'k'
```

**A result means nothing unless every setting is written down beside it. The hash names a row of the log.**

# The retriever: arc02, credit facility outstanding

**retrieve(question): three chunks, best first; the key ($160.7 million) is on page 37**

```
rank 1: boyd10k-7-036-0413, Item 7, page 36, score 0.6802  [boyd10k; Item 7; p. 36; c0413]
rank 2: boyd10k-1A-019-0225, Item 1A, page 19, score 0.6375  [boyd10k; Item 1A; p. 19; c0225]
rank 3: boyd10k-8-082-0917, Item 8, page 82, score 0.6273  [boyd10k; Item 8; p. 82; c0917]
```

**The top chunk, quoted as published: holds 160.7: True; the second chunk holds 0.2 billion: True**

```
 the Credit Facility for

2024.

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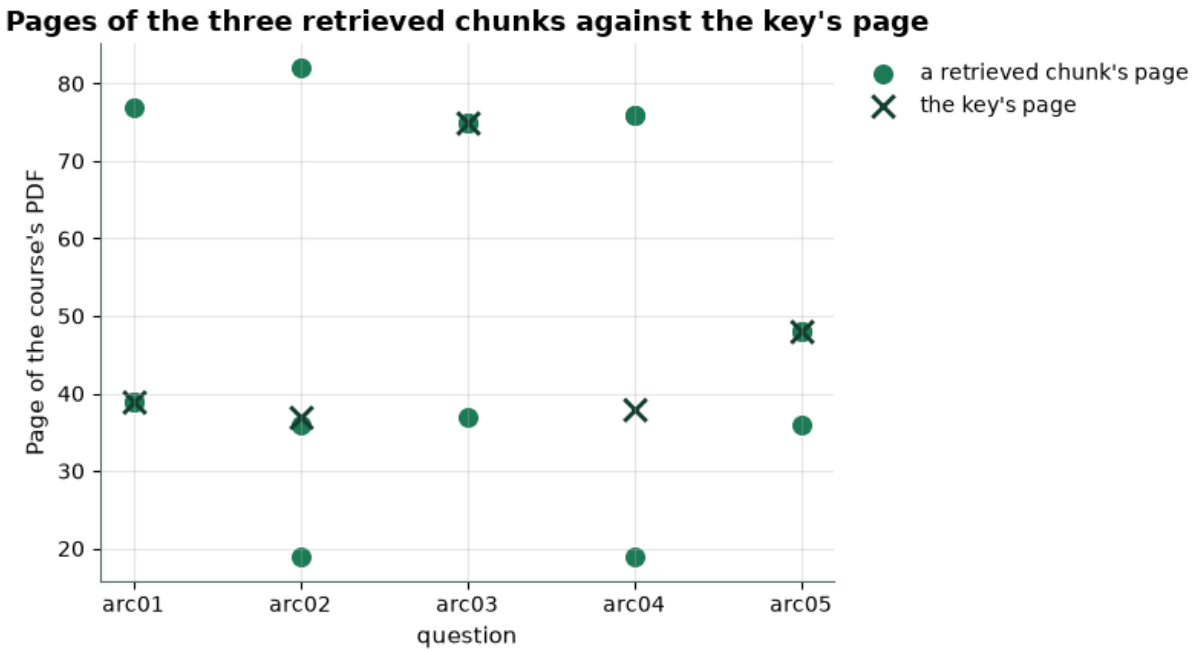
Indebtedness

The outstanding principal balances of long-term debt, before unamortized discounts and fees, and the changes ...
```

Source: Boyd Gaming Corporation, Form 10-K for fiscal 2025, filed 2026-02-20, accession 0001437749-26-004908, Item 7, page 36 of the course's PDF.

**The model is about to see both figures, 160.7 and 0.2 billion. The key is $160.7 million.**

# Three rules, three counts




**3 of 5**

by page

**4 of 5**

by quote

**4 of 5**

by figure

| id | key page | pages | page quote | figure | |
|---|---|---|---|---|---|
| arc01 | 39 | 39, 39, 77 | 1 | 1 | 1 |
| arc02 | 37 | 36, 19, 82 | 0 | 0 | 1 |
| arc03 | 75 | 75, 75, 37 | 1 | 1 | 1 |
| arc04 | 38 | 76, 19, 76 | 0 | 1 | 0 |
| arc05 | 48 | 48, 48, 36 | 1 | 1 | 1 |

**Settled today: a retrieval miss is counted by the figure rule; day 11's recall stays by page; every count names its rule.**

# The prompt: three messages

| | |
|---|---|
| **system** | Course 7's SYSTEM_PROMPT: exact quotes, section numbers, NOT IN TEXT, the refusal line |
| **system** | CITATION_RULE: labelled chunks; answer only from them; a label after every sentence; NOT IN CORPUS |
| **user** | Chunk 1: [label] text / Chunk 2 / Chunk 3 / Question: ... |

**3**

messages: system, system, user

**2 + 1**

two system, one user

**[509, 444, 1677]**

characters, arc02

**Nothing else goes in: no key quote, no page outside the chunks, no hint about the answer.**

# One answer: arc02 in the saved run

**answer(client, question, retrieve, settings, mode); key: $160.7 million**

```
Approximately $0.2 billion was outstanding under the Credit Facility as of December 31, 2025 [boyd10k; Item 1A; p. 19; c0225].
```

| 1 | no match | no | yes |
|---|---|---|---|
| label cited; sent: yes | grade against the key | states 160.7 | states 0.2 billion |

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 1 request, 700 prompt and 41 completion tokens; saved run recorded 2026-10-13; replayed (no key loaded).

```
chunker=recursive 512/64, embedder=minilm@1110a24, index=faiss exact, retrieve=dense k=3, rerank=none, expand=0, model=gpt-4.1-mini, temperature=0, top_p=1, max_tokens=300
```

**Grounded is not right: every word is in the text sent, the citation is real, and the key is missed.**

# The local route, on a T4

### LOCAL_MODEL, read on the Hub 2026-10-12

| | |
|---|---|
| **file** | `Mistral-7B-Instruct-v0.3-Q4_K_M.gguf` |
| **size** | `4.37 GB (the repository's file list)` |
| **licence** | `apache-2.0 (the card's metadata)` |
| **runner** | `llama-cpp-python 0.3.36, MIT` |
| **wheels** | `the project's CUDA index, cu118 to cu132` |
| **on a CPU** | `22.4 s for one answer, one thread` |

### What the saved notebook prints

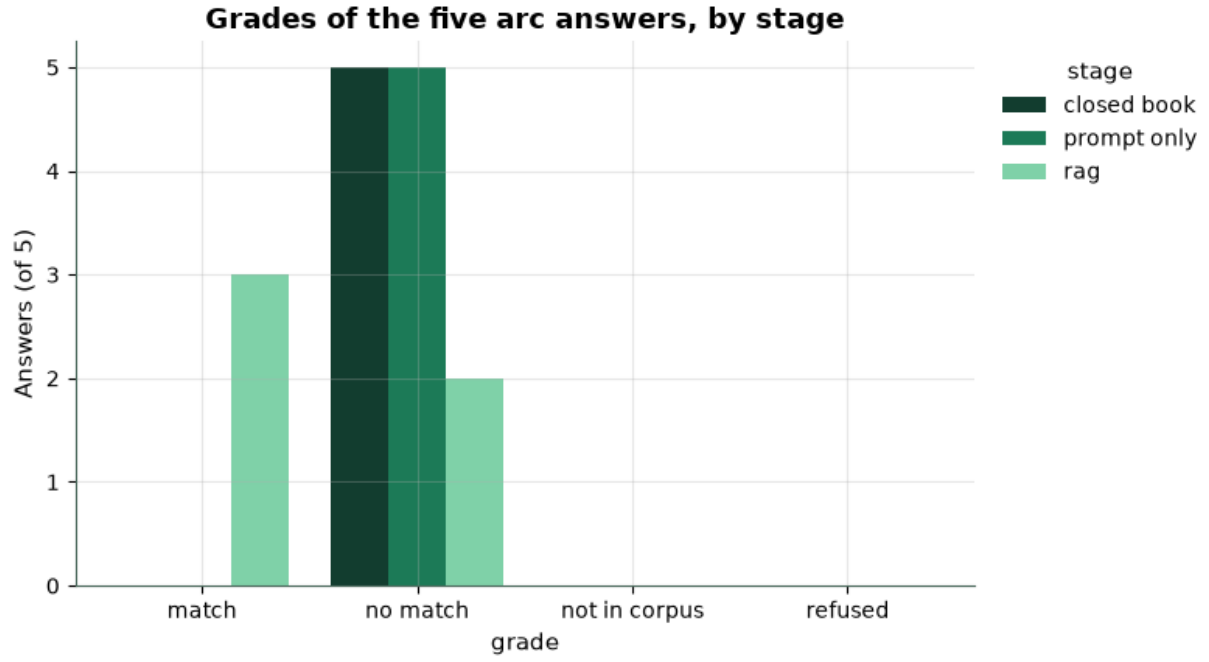
```
No GPU in this runtime: this cell is skipped. In
Colab, change the runtime to the T4 GPU and run the
notebook from the top.
```

```
GPU found: False. Local-model cells: skipped.
```

In Colab on a T4 the cell installs the wheel for the runtime's CUDA, downloads the file once, and answers arc02 from the same three chunks.

**GPU advised in Colab, never required; the recorded lesson runs on the API, inside the four-minute budget.**

# Fifteen answers, graded against their keys




**0 of 5**

closed book (day 1)

**0 of 5**

prompt only (day 1)

**3 of 5**

RAG (today)

### Citations

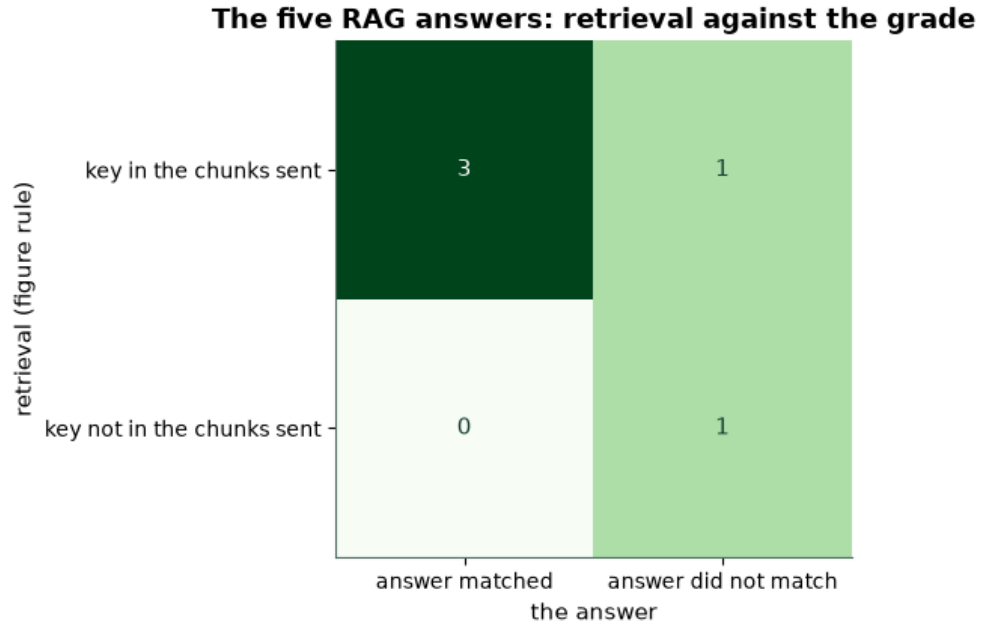
5 of 5 cite only chunks that were sent

3 of 5 end every sentence with a label

Two answers quote two sentences and cite once at the end: day 10's check will reject the uncited one.

**Closed book 0, prompt only 0, RAG 3 of 5: retrieval is where the chart changes.**

# Retrieval against the grade




## Two kinds of miss, by the figure rule

**retrieval miss: arc04**

The leverage ratio sat in a chunk that was not retrieved: no answer could have matched.

**generation miss: arc02**

160.7 in the top chunk, 0.2 billion in the second; the model took the second and cited it.

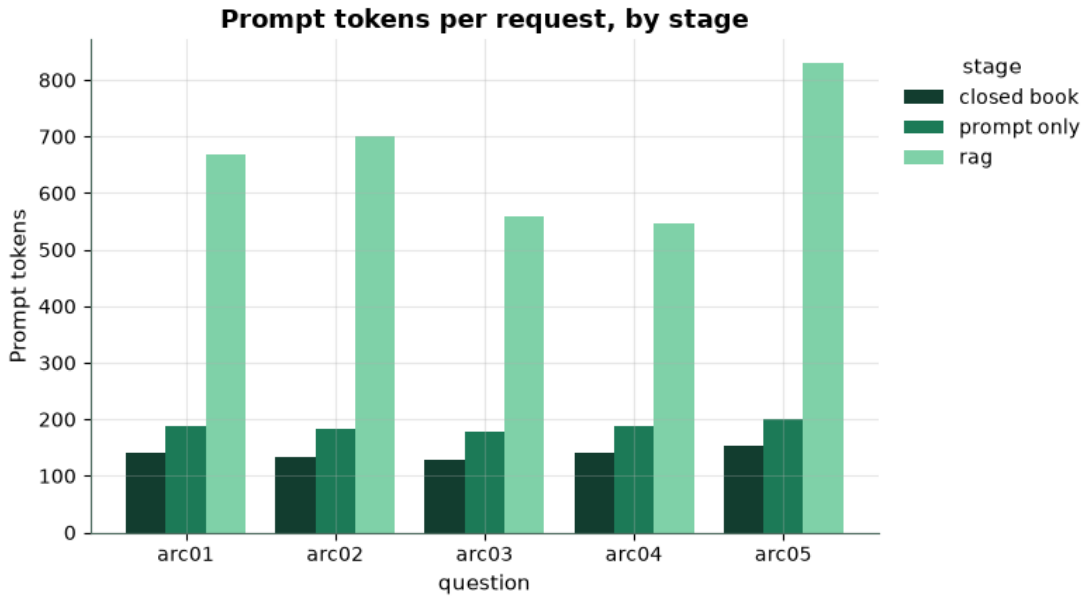
## The same five, by two rules

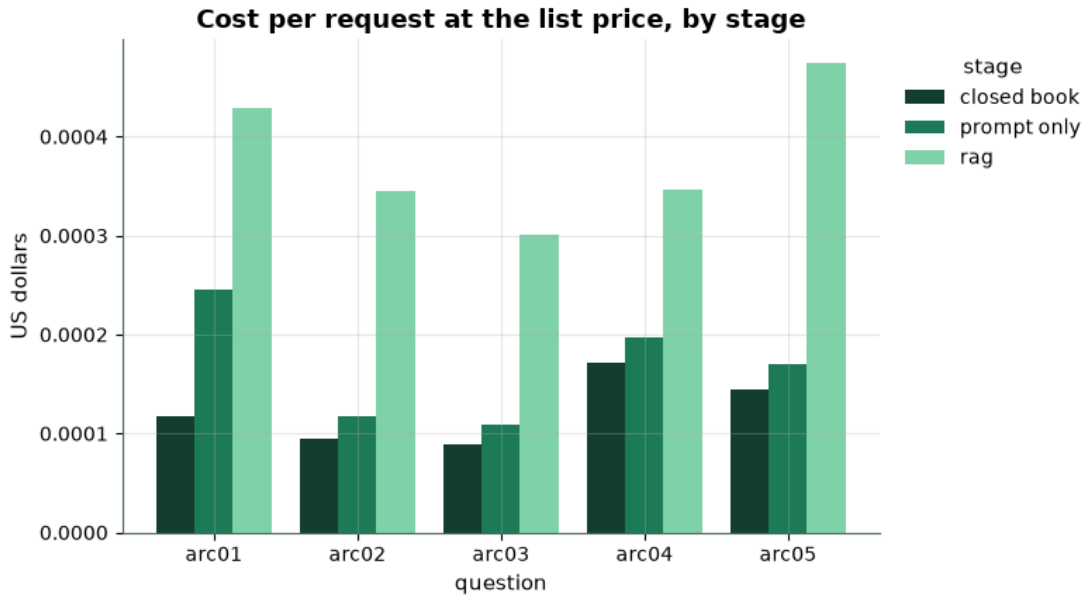
by page 3 of 5 retrieved; by figure 4 of 5

every citation names a chunk that was sent: 5 of 5

**Rows: the key's figures in the chunks sent. Columns: the answer matched. Read the off-diagonal cells.**

# Tokens and cost by stage




**stage: prompt tokens, completion tokens, USD, over five requests**

```
closed book     697   211 0.000616
prompt only     937   290 0.000839
rag            3305   359 0.001896
```

**Fifteen requests: 4,939 prompt and 860 completion tokens, US$0.003352; a RAG request carries 4.7 times the prompt tokens.**

# What the pipeline shows, and where it fails

### Finding 1, as counts

closed book 0 of 5, prompt only 0 of 5, retrieval 3 of 5 matched their keys, on these five questions and this document

### Finding 2, as counts

of 2 RAG misses, 1 came from retrieval (the key's figures were not in the chunks sent) and 1 from generation (they were sent and the answer did not match)

## 3 / 4 / 4

of 5 retrieved, by page / quote / figure

## one key

match = the key's figures in the answer

## one document

as of one filing date

# Excel side by side, and Git

### Excel side by side (tested in Excel)

| | |
|---|---|
| `=COUNTIFS($B$2:$B$16,J2,$E$2:$E$16,"match")` | 0, 0, 3 by stage = groupby |
| `=COUNTIFS(E2:E6,1,G2:G6,1)` | 3: key in the chunks and matched |
| `=COUNTIFS(E2:E6,1,G2:G6,0)` | 1: key in the chunks, not matched |
| `=COUNTIFS(E2:E6,0,G2:G6,0)` | 1: key not in the chunks, not matched |
| `=COUNTIF(G2:G6,1)/COUNTA(G2:G6)` | 0.6000 matched share |
| `dtype=str; the question and pages cells` | 15 of 15 ids read back as text |

```
run,settings_hash,questions,recall_at_k,answered,matched_key,grounded,mean_prompt_tokens,usd
day05_rag_k3,6579143b53aa799442ea4d46869c3821681c5811d4671c546cba697f10cd0b09,5,0.6000,1.0000,0.6000,1.0000,661.0000,0.0019
```

**Git: rag_results.csv, one row named by the settings hash; nine files committed; tag week1; 32 tests.**

# The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

## Docstring

Units, conventions, and what the code must never let through: what you asked for.

→

2

## Code back

Written for the course in the style of an AI answer. Read it.

→

3

## Test first

Plant the case the docstring promises to catch. Run it: it fails.

→

4

## Fix

Find the rule it broke, fix it, run the test again.

**Today: the chunks go after the question with no labels, so an answer has nothing to cite.**

*Code that runs is not the same as code that is right.*

# What you can do now

- Every setting in one frozen record, printed as a settings line
- A retriever; hits counted by page, by quote, and by figure
- The prompt: system, citation rule, labelled chunks, question
- One answer, read with its LLM line and its settings line
- A retrieval miss told from a generation miss

**Week 2, Day 1** BM25 and hybrid search